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324614 (24)

BE (6th Semester)

Examination, April-May, 2014

Branch : Electrical

MICROPROCESSORS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Answer any two parts from the remaining three.

Q. 1. (a) Why are low order address bus & data bus multiplexed in 8085 microprocessor ? Which signal is used to demultiplex the AD bus in 8085 ? 2

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(b) How many 16 bit registers are there in 8085 microprocessor ? Explain the significance of each. 7

(c) Differentiate between : 7

(i) LXI H, 4455 H & LHLD 4455 H

(ii) JMP CO50 H & CALL CO50 H

(iii) CPI 52 H & SUI 52 H

Also give the addressing modes, Number of bytes for each instruction.

(d) Write a program to multiply two 8 bit numbers stored at 3000 H & 3001 H memory locations. Store the result at two consecutive memory locations. Write the program in the following format : 7

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Label, Address Mnemonics Comments.

Q. 2. (a) How many address lines are required to address one mega byte memory ? 2

(b) Compare static & dynamic RAM giving examples of each. 7

(c) Define write cycle time of 6264 static RAM chip. Explain the write cycle of 6264 RAM chip. Indicate write cycle times in the diagram. 7

(d) Describe the EPROM programming procedure. 7

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Q. 3. (a) If an I/O device is an I/O mapped I/O which instructions are used for data transfer with the I/O device ? 2

(b) Give the classification of data transfer schemes. Explain in brief the interrupt driven data transfer. 7

(c) Explain in brief the instructions used to handle interrupts. 7

(d) Explain how bus contention can be avoided using 2-line control. 7

Q. 4. (a) What do you mean by an I/O port ? 2

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(b) Explain the signals of Intel 8212, an 8 bit I/O port. Explain the input mode with a neat schematic diagram. 7

(c) Draw the block diagram of the 8259, programmable interrupt controller and explain the significance of each block. 7

(d) Explain how AD-7820 is interfaced to 8085 microprocessor. 7

Q. 5. (a) Enlist the software development aids of a microcomputer development system. 2

(b) Explain in brief the constituents of an operating system. 7

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Write short notes on single board computer system. 7

(d) Define instruction cycle. Draw the timing diagram of memory write machine cycle. 7

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